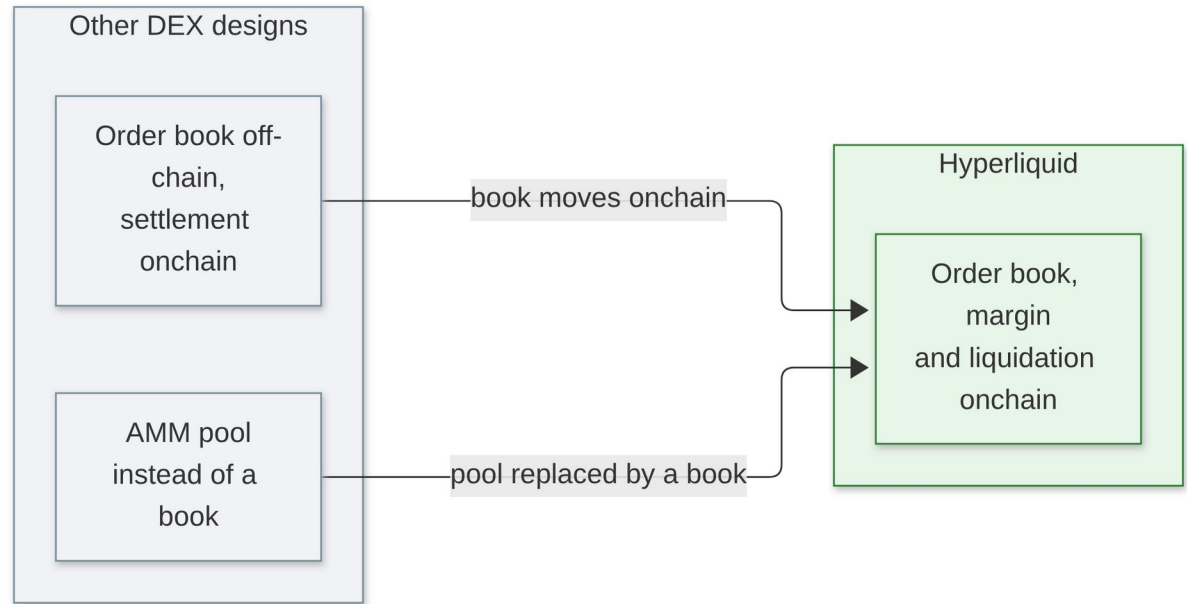


Hyperliquid in Five Minutes

An L1 where the order book is part of consensus
Ryan Sauge - v1.0 - 2026-10-01

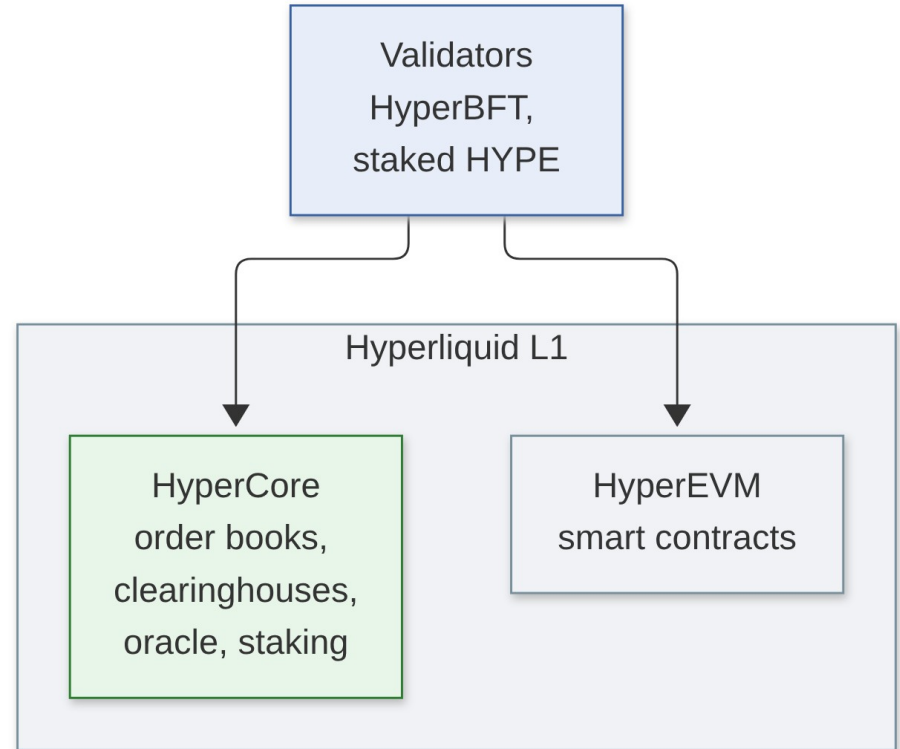
What is Hyperliquid

- A DEX for perpetual futures and spot, on its own chain
- Many DEXs keep the book off-chain, or use an AMM
- Here, order book, margin and liquidations are onchain state
- Question: what if the exchange is the chain itself?

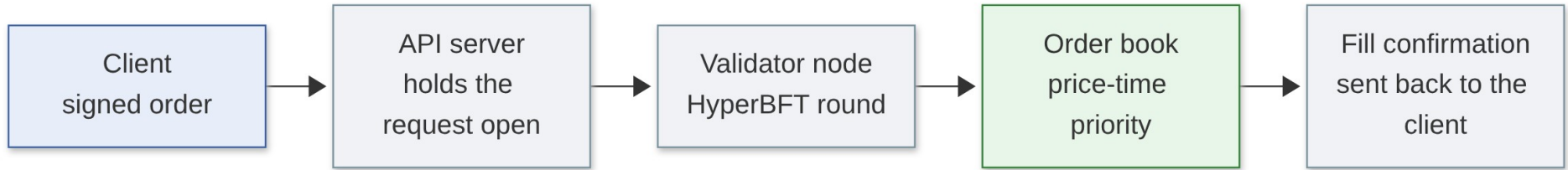


Architecture: one consensus

- HyperBFT: a block is final with 2/3 of staked HYPE
- HyperCore: order books, clearinghouses, oracle, staking
- HyperEVM: a regular EVM, chain ID 999
- Same consensus, no bridge between the two

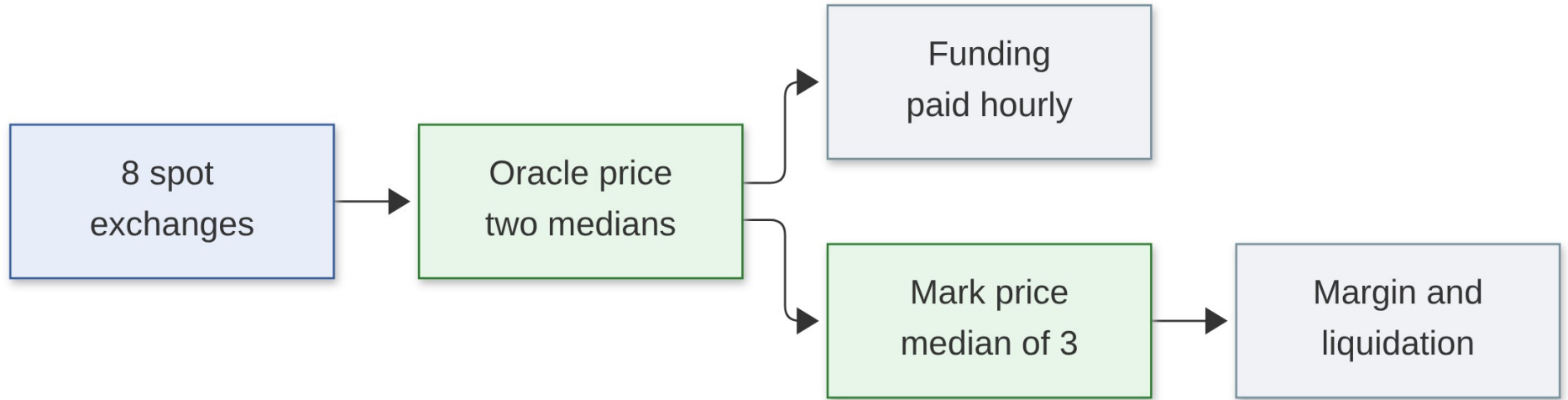


The life of an order



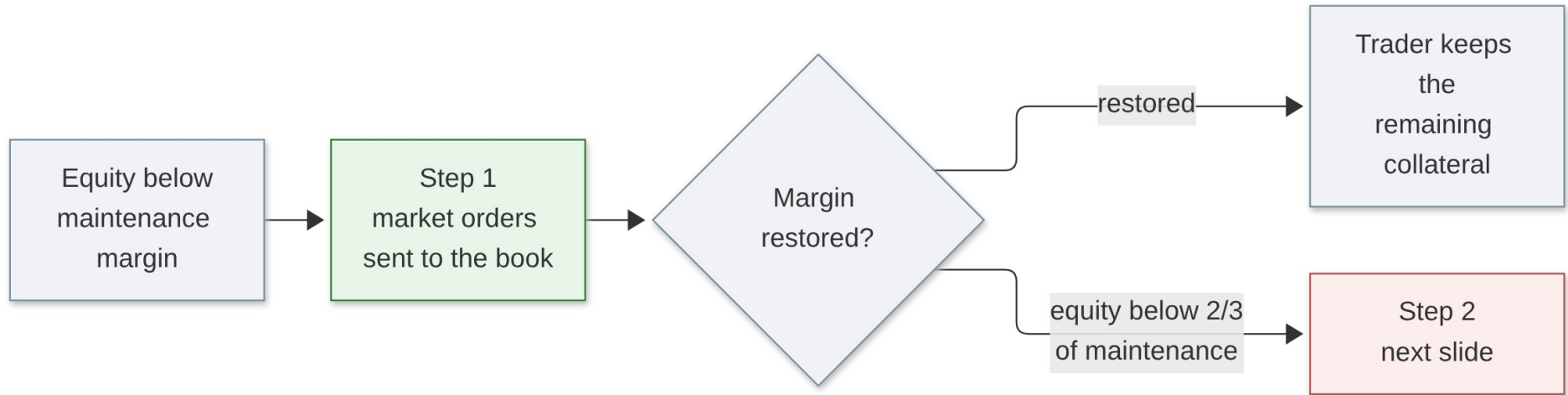
- Inside a block, actions run in three groups:
 - Orders that take no liquidity (post-only)
 - Cancels
 - Orders that take liquidity (GTC, IOC)
- Margin is checked at placement and again at each match

Prices: oracle and mark



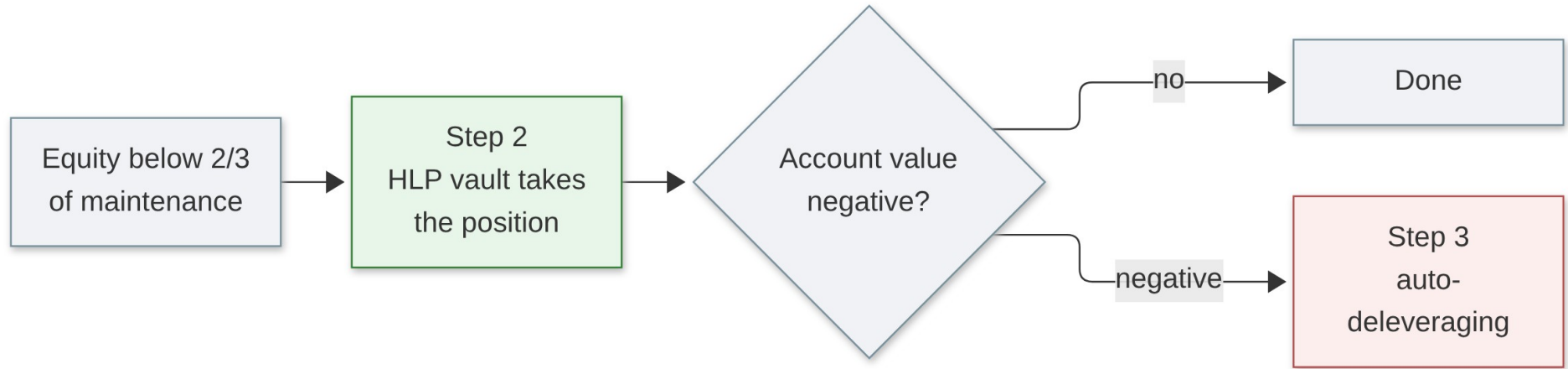
- Oracle: two medians, every ~3 s, used for funding
- Mark: median of 3 indices, used for margin and liquidation

Liquidation: on the order book



- Anyone can take the other side of the liquidation orders
- Above 100 000 USDC: 20% at a time, then a 30 s cooldown

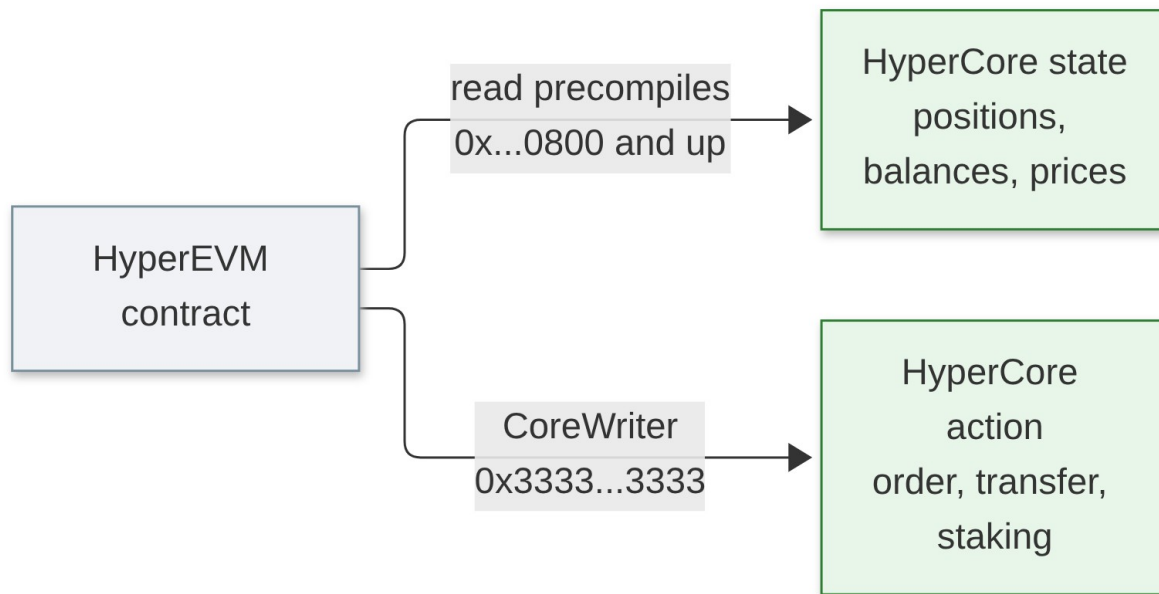
Liquidation: backstop and ADL



- Auto-deleveraging (ADL) closes the most profitable, most levered opposite positions first
- A user with no open position never pays for platform losses

HyperEVM and HyperCore

- Read: values match HyperCore when the EVM block was built
- Write: CoreWriter orders are delayed by a few seconds
- Small blocks every second (3M gas)
- Big blocks every minute (30M gas)

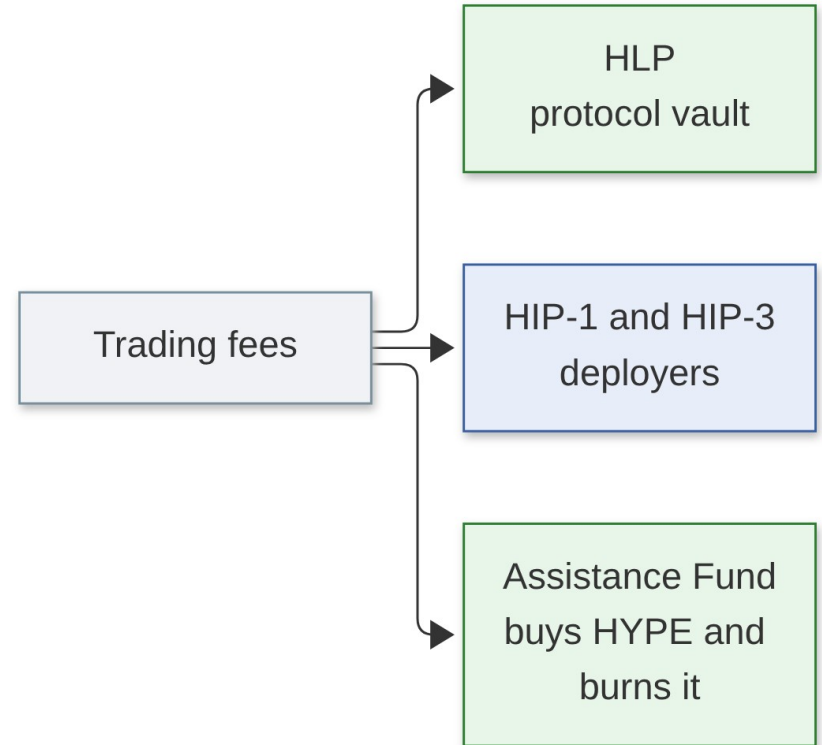


The HIP standards

Standard	What it adds
HIP-1	Anyone can deploy a token and its spot market (Dutch auction, 500 HYPE floor)
HIP-2	Hyperliquidity: onchain market making, 0.3% spread refreshed every 3 s
HIP-3	Anyone can run a perp exchange; the deployer publishes the oracle, 500 000 HYPE staked
HIP-4	Fully collateralised outcome markets, no leverage, e.g. prediction markets

Where the fees go

- Base perp fees: 0.045% taker, 0.015% maker
- Staking HYPE: up to 40% off
- No operator cut; HLP also runs backstop liquidations
- Staking rewards: ~2.37%/year at 400M HYPE staked



Limitations

Limitation (by design)

What it means for a user

2/3 of the stake can change any state

Order book, oracle and votes share one trust assumption

Validator votes decide offchain questions

Delistings, HIP-3 slashing, quote assets: judgement by stake

HIP-3 oracle is one deployer

Slashed stake is burned; users are not compensated

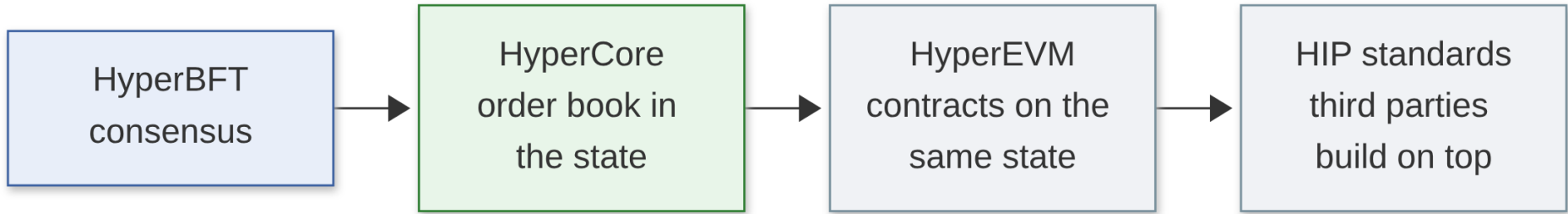
HyperEVM token links are not verified

A linked contract can hold arbitrary code

No automatic slashing of validators

Misbehaving validators are jailed by vote

Conclusion



- On Hyperliquid, the exchange is the chain's state machine
- Cancels before takers, margin checked at each match, fixed liquidation steps
- The cost: one trust assumption, two thirds of the staked HYPE